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OCC SOFTWARE ENGINEERING Y2

SDT621: PM04

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Declaration Of Authenticity

I Fernando Laas (FULL NAME)

hereby declare that the contents of this assessment SDT621 PM04

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

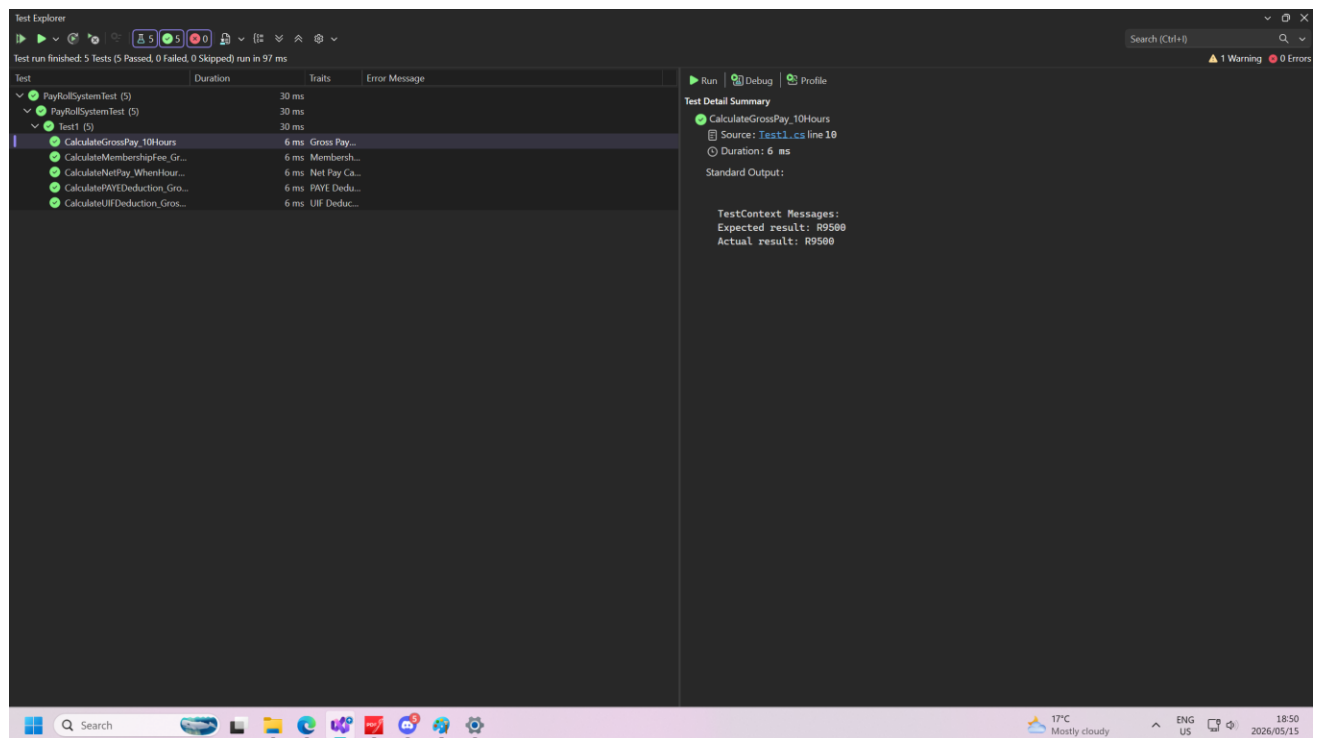
Signature: 

Date: 15/05/2026

Unit Testing

Test Case	Input	Expected Output	Result
Gross Pay	Hours worked = 10	9500	9500
UIF	Gross pay = 10000	100	100
Membership	Gross pay = 10000	1300	1300
PAYE	Gross pay = 10000 Dependents = 2	2212.5	2212.5
Net Pay	Hours worked = 10 Dependents = 2	6068.125	6068.125

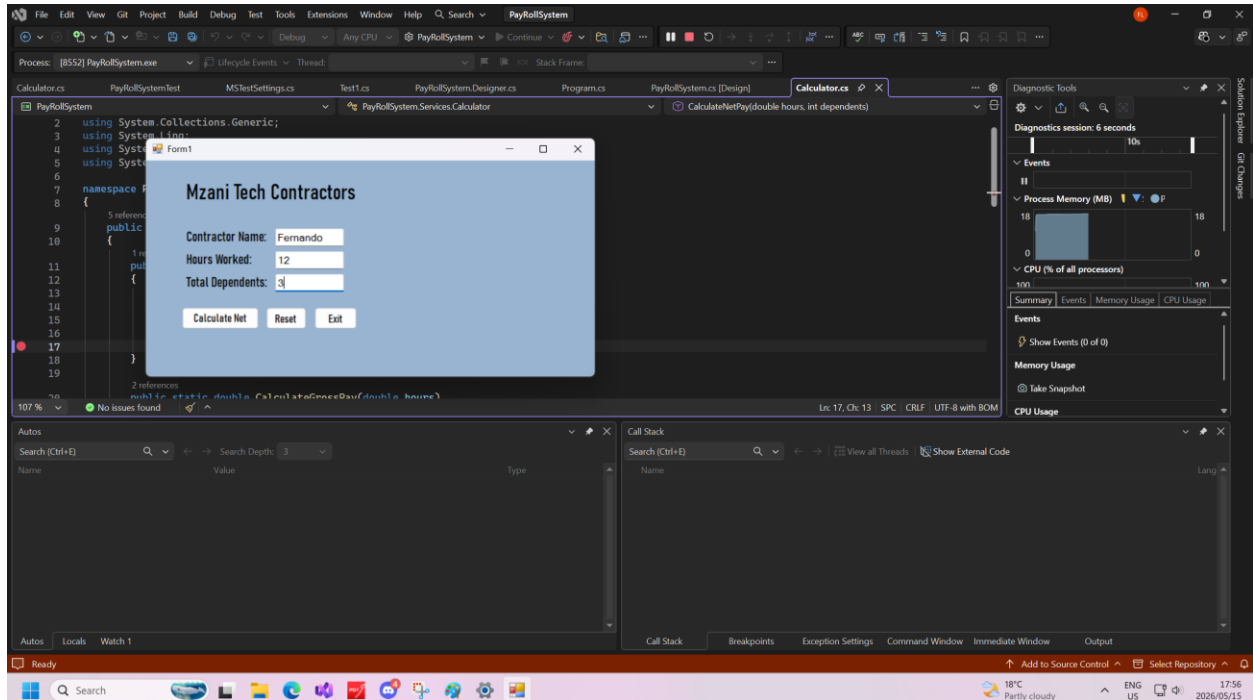
Test Explorer:



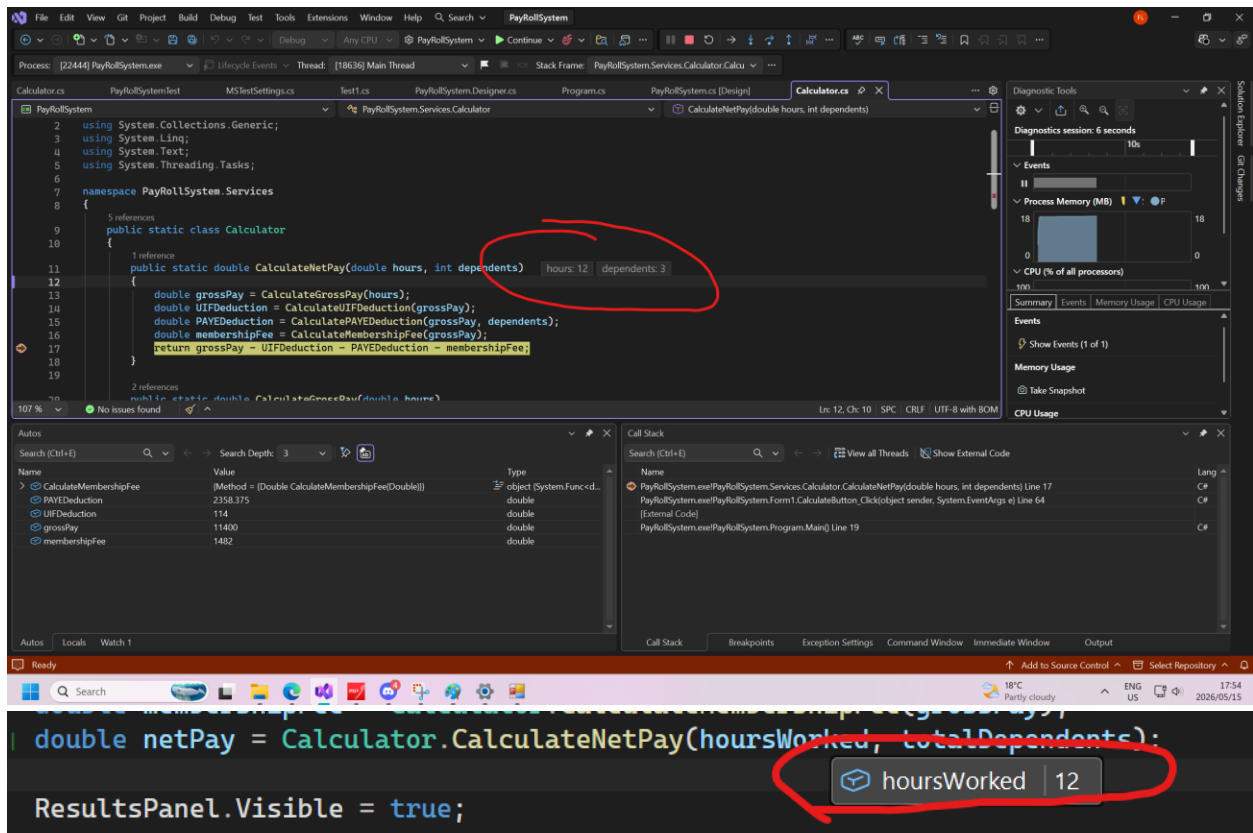
Integration Testing

Breakpoints were used to ensure that all components of the system were interacting and working together correctly.

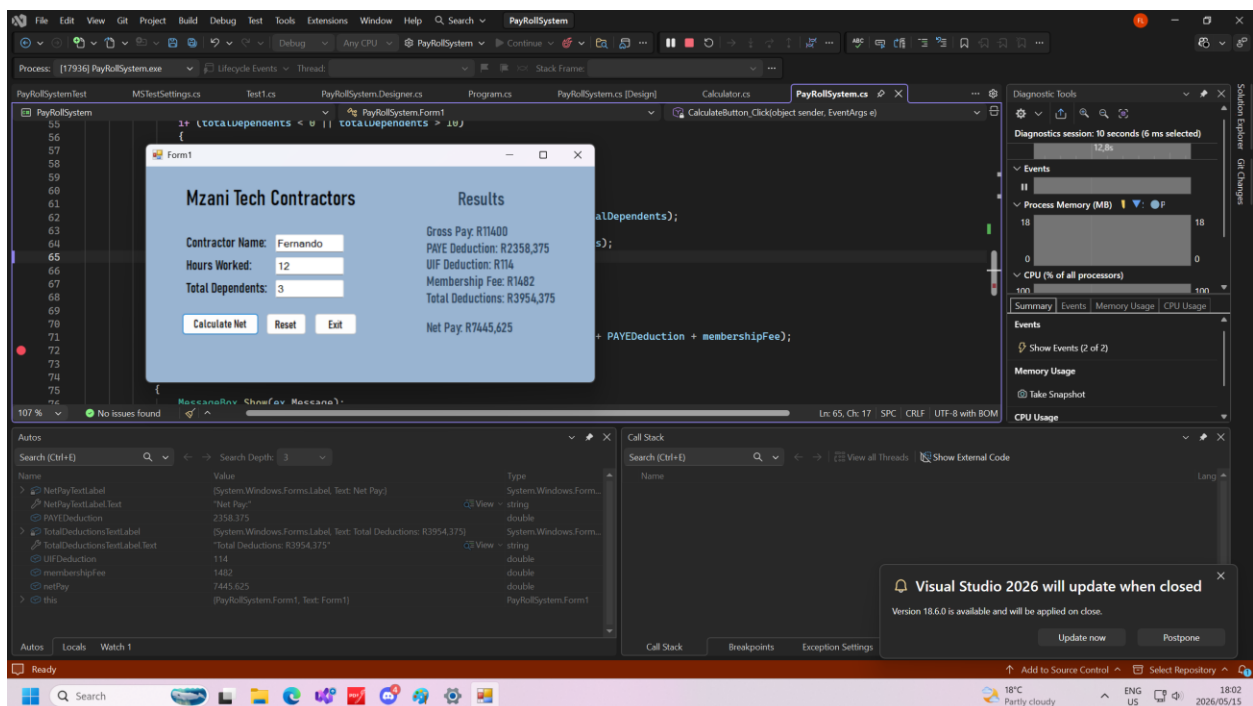
Input:



Calculation:



Output:



System Testing

Step	Action	Expected Result
1	Pressing the 'calculate' button with valid input	Display results
2	Pressing the 'calculate' button with invalid input	Display message box with relevant error message
3	Pressing the 'calculate' button with no input	Display message box with relevant error message
4	Pressing the 'clear' button	Empties the input fields and hides the 'results' section
5	Pressing the 'exit' button	Exits the application

Retesting

After fixing a bug related to the calculation of the PAYE deduction, the system was retested. After retesting the system, the system was confirmed to be fully functional and without defects.

Regression Testing

Action	Expected Result	Actual Result	Defects Found	Corrections Made
Pressing the 'calculate' button with valid input	Display Results	Display inaccurate results	Typo in calculations	Adjusted the calculations to use double instead of int
Pressing the 'calculate' button with invalid input	Display message box with relevant error message	Display message box with relevant error message	None	N/A
Pressing the 'calculate' button with no input	Display message box with relevant error message	Display message box with relevant error message	None	N/A
Pressing the 'clear' button	Empties the input fields and hides the 'results' section	Empties the input fields and hides the 'results' section	None	N/A
Pressing the 'exit' button	Exits the application	Exits the application	None	N/A

After all testing and troubleshooting steps, the system is free of issues and remains in a usable state.